PROJECT PLANNING PHASE
Project Name: Plugging into the Future:
An Exploration of Electricity Consumption Patterns

Date	01 JULY 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	5 Marks

Introduction

The Project Planning Phase focuses on organizing the activities, resources, schedule, and milestones required for the successful completion of the Electricity Consumption Analysis project. It ensures that every stage of the project is completed systematically within the planned timeline while maintaining quality, proper documentation, and timely delivery.

Objectives

- Design an efficient electricity consumption analysis system.
- Develop an interactive Tableau dashboard and story.
- Store and manage electricity data using MySQL.
- Provide meaningful business insights through data visualization.
- Integrate the dashboard into a Flask web application.
- Support future enhancements and scalability.

Project Planning

Project planning is an essential phase that defines the activities, resources, and timeline required for the successful completion of the Electricity Consumption Analysis project.

The project was divided into multiple phases to ensure systematic development, proper documentation, timely completion, and successful deployment. The planning process included collecting the dataset, storing it in MySQL, performing SQL analysis, creating Tableau visualizations, publishing the dashboard on Tableau Public, integrating it with a Flask web application, and preparing the final documentation.

A structured weekly schedule was followed to complete all project tasks efficiently.

Week	Activity
Week 1	Project topic selection and problem identification
Week 2	Dataset collection and preprocessing
Week 3	MySQL database creation and SQL query execution
Week 4	Tableau visualization development
Week 5	Dashboard and Story creation

Week 6	Flask web application development and Tableau Public publishing
Week 7	Testing, GitHub repository creation, and project validation
Week 8	Final documentation, report preparation, and project submission

Project Planning Components

The Project planning Phase consists of the following components:

1. Proposed Solution

Describes the overall solution for analyzing electricity consumption using MySQL, Tableau Desktop, Tableau Public, and Flask. It explains how the system addresses the identified problem and delivers interactive dashboards for effective analysis.

2. Solution Architecture

Defines the overall structure of the system, including data collection, database management, visualization, dashboard publishing, and web application deployment. It illustrates the interaction between all system components.

3. Workflow

Describes the step-by-step process followed in the project, starting from dataset collection and ending with dashboard visualization and decision-making.

4. Technology Integration

Explains how different technologies such as **Excel**, **MySQL**, **SQL**, **Tableau Desktop**, **Tableau Public**, **Flask**, **Python**, and **GitHub** work together to build the complete analytics solution.

Design Goals

- Develop a reliable and scalable system.
- Ensure efficient data processing.
- Provide interactive visualizations.
- Improve user experience.
- Enable easy deployment through a web application.

Risk Management

The following potential risks were identified during the project planning phase:

- Missing or inconsistent electricity consumption data.
- SQL query execution errors.
- Difficulty in connecting Tableau with MySQL.
- Dashboard design and visualization challenges.
- Tableau Public publishing issues.
- Flask integration and deployment issues.
- Delays in report preparation and documentation.

Risk Mitigation

The identified risks were minimized through:

- Dataset validation and preprocessing.
- Regular SQL query testing.
- Verifying MySQL database connectivity.
- Continuous dashboard testing in Tableau Desktop.
- Publishing verification on Tableau Public.
- Testing Flask application locally before deployment.
- Maintaining backups of project files.
- Following a structured project timeline.

Deliverables

The project planning phase identified the following deliverables:

1. Cleaned Electricity Consumption dataset.
2. MySQL Database (electricity).
3. SQL Query File (queries.sql).
4. Tableau Workbook (Electricity_Consumption_Analysis.twb).
5. Interactive Dashboard.
6. Tableau Story with Business Insights.
7. Tableau Public Dashboard Link.
8. Flask Web Application.
9. GitHub Repository.
10. Final Project Report.
11. Performance Testing Report.

Milestones Achieved

- ✓ Dataset successfully collected and imported.
- ✓ MySQL database created.
- ✓ SQL queries executed successfully.
- ✓ Five Tableau worksheets created.
- ✓ Interactive dashboard developed.
- ✓ Tableau Story completed.
- ✓ Dashboard published on Tableau Public.
- ✓ Flask web application integrated successfully.
- ✓ GitHub repository created.
- ✓ Project documentation completed.

Expected Outcome

The Project Planning Phase results in a well-organized project schedule, efficient resource utilization, timely completion of activities, successful implementation, and proper documentation. It ensures that the Electricity Consumption Analysis project is completed systematically while achieving all planned objectives and deliverables.

Conclusion

The Project Planning Phase provides a structured roadmap for successfully completing the Electricity Consumption Analysis project. Through proper scheduling, risk management, milestone tracking, and resource planning, the project can be completed efficiently within the planned timeline. Effective project planning contributes to better coordination, successful implementation, and high-quality project outcomes.